

The PCB DESIGN ECOSYSTEM In India Is Not Supported Adequately By HARDWARE PROTOTYPING

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With the increasing role of electronics in everyday life, the need for a reliable PCB supplier is critical. C. Krishna Rao, chairman; Dr V. Venkata Ramani, director - business development; and S. Rambabu, general manager - commercial and technical, Sulakshana Circuits Ltd, Hyderabad, throw light on the same, in an interaction with Baishakhi Dutta

Q. How does the choice of a PCB manufacturer affect the Internet of Things (IoT) and embedded systems projects?

A. For the IoT and embedded systems an excellent PCB manufacturer is important because time-to-market is critical for success in this quickly changing field. Normal bare-board testers (BBTs) used for volume production delay prototype testing.

Second, a good PCB manufacturer should carefully check design rules and ensure manufacturability of designs.

Finally, with the small size of IoT devices, excellent alignment of the routed profile of the board and the electrical circuit is critical. CNC drilling and routing machines are critical to maintain alignment.

Q. Does India have a strong base of players who can create innovatively?

A. Absolutely!

Q. Is the PCB design ecosystem mature?

A. It has certainly come a long way. Indian designs are typically simpler than those in the US—fewer layers, wider lines, etc. There are only a few companies since much of our electronics come from China. And, many engineers still prefer careers in software rather than hardware.

The PCB design ecosystem in India is not supported adequately by hardware prototyping. Often, Indian designs are tested with prototyping in China.

Q. What quality considerations do you keep in mind while creating PCBs?

A. We triple-check designs before manufacturing. We also make sure that these are manufacturable, that is, design rule checks like solder mask clearances around holes are adequate, track-to-track, track-to-pad clearances are maintained and so on. Most problems

are fixed using computer-aided manufacturing (CAM) software. But if there is still an issue, the designer is consulted.

Q. What kind of EDA software do you use for designing PCBs?

A. We do not use EDA software—that is a designer's tool. We use CAM.

Q. Is India mature enough to deploy the dark factory concept?

A. No, because here capital is expensive, while labour is cheap. Also, scale and the inherent complexity of PCB manufacturing is not conducive to expensive automation.

Q. What technologies are you using to bring in quick-turn capabilities?

A. For quick-turn capabilities, we have electronic files that control many of our machines derived from customer-provided electronic data. On receiving customer data, we verify the files and develop programs for CNC drills, CNC routers and BBTs, and plot films. This pre-production engineering is done digitally.

Q. What challenges do you face as an electronics manufacturing business in India?

A. We are fighting Chinese competition with one hand tied behind our back. Issues like poor water supply, intermittent electricity, poor roads, expensive financing, slow customs and logistics, and a poor duty structure are some challenges.

Q. Have you sought help from the government?

A. Differential duty between copper clad laminates, bare and assembled PCBs, and final products to encourage manufacturing has been communicated to the government many times. Phased manufacturing programmes to systematically develop specific large verticals such

as mobiles, energy meters, set-top boxes, computers, etc would also be helpful.

The government must insist that its orders for PCB purchases are sourced in India.

Q. Where do you see India's hardware manufacturing capability today?

A. India has the technical capabilities but not scale. While her software and pharmaceutical industries are world class, the hardware industry is far below the necessary scale. As a result, the ecosystem does not exist, and long supply chains are needed.

Q. Any message for the government to improve India's electronics manufacturing ecosystem?

A. Slightly higher duties on bare and assembled PCBs, assembly and box build will push more manufacturing into India. That would drive more electronic inventories being maintained here, along with more design services (manufacturing and design engineering). There will be many more jobs created this way than from chasing chip factories.

Q. Would you recommend automation to other manufacturers in India?

A. Yes! The only way to ensure quality, quicker deliveries and lower costs is to implement as much automation as possible.

Q. What are your three golden rules for electronics design engineers?

A. My top three rules are:

1. Carefully complete designs including EMI checks, thermal analysis, etc.

2. Do not cut corners on clearances in tight areas to get everything to fit.

3. Show PCB designs early to the PCB manufacturer and parts' suppliers, and take their advice on cost reduction. **EFY**